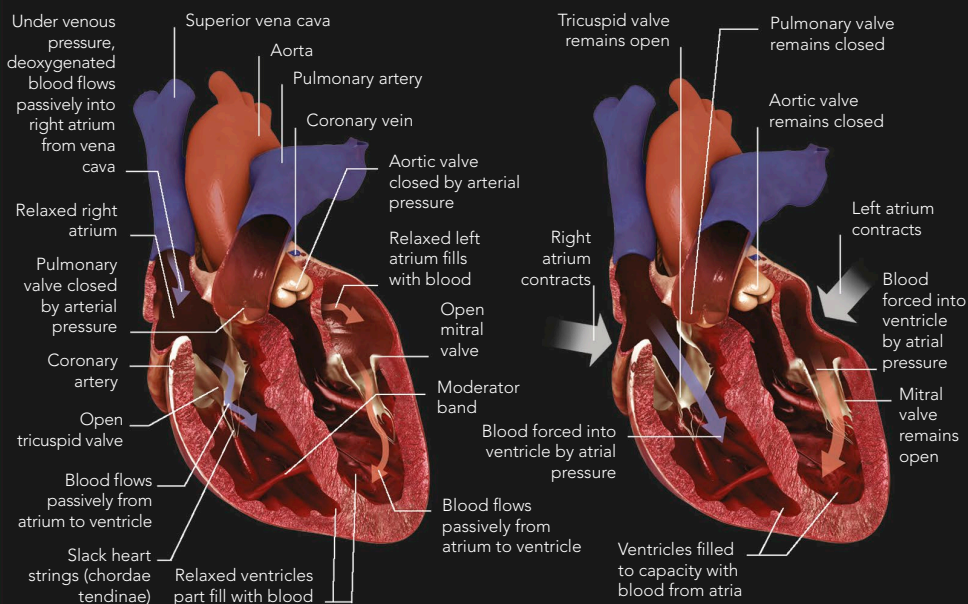


HOW THE HEART BEATS

THE HEART IS A DYNAMIC, UNTIRING, PRECISELY ADJUSTABLE DOUBLE-PUMP THAT FORCES BLOOD AROUND THE BODY'S IMMENSE NETWORK OF BLOOD VESSELS – PERHAPS MORE THAN THREE BILLION TIMES DURING A LIFETIME.

The heart's two lower chambers (ventricles) have thick, muscular walls that contract to squeeze blood into the arteries. The upper chambers (atria) have thinner walls and act partly as reservoirs for blood entering from the main veins. Each heartbeat has two main phases: in the first phase (diastole),

the heart relaxes and refills with blood; in the second phase (systole), it contracts, forcing the blood out. The whole cycle takes, on average, less than a second. During activity or stress, both the beating rate and the volume of blood pumped out of the heart increase greatly.



1 RELAXATION (LATE DIASTOLE)

During this phase of the heartbeat sequence, the muscular walls of the heart relax. The atrial chambers balloon slightly as they fill with blood coming in under quite low pressure from the main veins. Deoxygenated blood from the body enters the right atrium, while oxygenated blood from the lungs enters the left atrium. Some of the blood in the atria flows down into the ventricles. By the end of this phase, the ventricles are filled to about 80 per cent of capacity.

2 CONTRACTION OF THE ATRIA (ATRIAL SYSTOLE)

The heart's natural pacemaker, known as the sinoatrial node, is located in the upper part of the right atrium. It "fires" electrical impulses, similar to those generated by nerves, which set off the contraction phase. Some impulses spread through the atrial walls and stimulate their cardiac muscle to contract. This contraction squeezes the blood inside the atria through the atrioventricular (tricuspid and mitral) valves into the ventricles, whose walls remain relaxed.